Prop Let G l.c.t.d.s.c. group, $(C_i)_{i\in I}$ directed family of closed subgroups of G s.t. $N_G(C_i)$ open $\forall i \in I$ and $G = \overline{\bigcup_{i\in I} C_i}$. If each C_i is elementary $\Rightarrow$ G is elementary with

$$\xi(G) = \sup_{i\in I}{}^{+} \xi(C_i)$$

↳ (stronger than last version)

↳ $\sup^{+}(\alpha) = \begin{cases} \alpha & \text{if } \alpha \text{ is succ} \\ \alpha+1 & \text{if } \alpha \text{ is limit} \end{cases}$

Lemma:
Let G be a t.d l c s.c. group and α be an ordinal. Then $\{N \trianglelefteq G \mid N \text{ is closed and } \xi(G/N) \leq \alpha\}$ is a filtering family.

↳ Proof: It suffices to show that

$$\Sigma := \{N \trianglelefteq G \mid N \text{ is closed and } \xi(G/N) \leq \alpha\}$$

is closed under pairwise intersection. Taking $M, N \in \Sigma$, the diagonal map $G \to G/N \times G/M$ induces an injective, continuous map $G/(N\cap M) \hookrightarrow G/N \times G/M$. Because $\alpha(G) \geq \alpha(G/N)$ and $\alpha(G/N \times G/M) = \sup\{\alpha(G/N), \alpha(G/M)\}$ we get

$$\xi(G/N\cap M) \leq \xi(G/N \times G/M) \leq \alpha \Rightarrow N \cap M \in \Sigma \;\blacksquare$$

Prop: Let G be a t.d.l.c.s.c. group and $\mathcal{R}$ be a family of closed normal subgroups of G with trivial intersection. Then

$$\xi(G) = \sup_{K\in\mathcal{R}}{}^{+} \xi(G/K).$$

If G is compactly generated and elementary, then there is $K\in\mathcal{R}$ s.t. $\xi(G) = \xi(G/K)$.

↳ Proof:

Every $K \in \mathcal{R}$ has rank less or equal to $\sup^{+}_{K\in\mathcal{R}} \xi(G/K)$. Applying the last lemma, we may expand $\mathcal{R}$ to a filtering family $\tilde{\mathcal{R}}$ s.t. $\sup^{+}_{K\in\tilde{\mathcal{R}}} \xi(G/K) = \sup^{+}_{K\in\mathcal{R}} (G/K)$. Replacing $\mathcal{R}$ with $\tilde{\mathcal{R}}$ we may assume, w.l.o.g., that $\mathcal{R}$ is a filtering family.

If G/K is non-elementary for some $K\in\mathcal{R} \Rightarrow G$ is also non-elementary and there is nothing to prove.

We thus assume G/K elementary $\forall K\in\mathcal{R} \Rightarrow G$ is also elementary. We also have $\xi(G/K) \leq \xi(G)$ $\forall K\in\mathcal{R}$, and it follows that

$$\sup_{K\in\mathcal{R}}{}^{+} \xi(G/K) \leq \xi(G).$$

Conversely, say that $\alpha+1 = \sup^{+}_{K\in R} G/K$. We argue that $\mathcal{G}(G)\leq \alpha+1$ by induction on α. The base case is trivial, since in this case $G=\{1\}$. Suppose α is a successor ordinal; say $\alpha=\beta+1$. For any compactly generated open subgrp $O\leq G$ and $K\in R$, the rank of $O/(O\cap K)$ is at most $\beta+2$, so its residual $\mathrm{Res}(O/O\cap K)$ has rank at most $\beta+1$. There is an injective, cont map from $\mathrm{Res}(O)/(\mathrm{Res}(O)\cap K)$ into $\mathrm{Res}(O/(O\cap K))$ then

$$\mathcal{G}(\mathrm{Res}(O)/(\mathrm{Res}(O)\cap K))\leq \beta+1.$$

$\mathrm{Res}(O/(O\cap K)) =$
$= \bigcap\{O \trianglelefteq_o O/(O\cap K)\}$
$\mathrm{Res}(O)/(\mathrm{Res}(O)\cap K) =$
$= \bigcap\{O \trianglelefteq_o O\}$
$K\cap(\bigcap\{O \trianglelefteq_o O\})$

Since $\beta<\alpha$ and β does not depend on K, the inductive hypothesis ensures that $\mathcal{G}(\mathrm{Res}(O))\leq \beta+1=\alpha$. It now follows that $\mathcal{G}(G)\leq\beta+2=\alpha+1$, since O is an arbitrary compactly generated open subgroup.

Let us now consider the case when α is an limit ordinal, and since $\alpha>0$, the ordinal α is transfinite. Take $O\leq G$ a compactly generated subgroup. If $\mathcal{G}(O)$ is finite, then the induction hyp. implies that $\mathcal{G}(O)=\sup^{+}_{K\in R}(\mathcal{G}(O/O\cap K))$.

Since $\{\xi(Q/Q\cap K)\}_{K\in\mathcal{R}}$ is a bounded collection of finite ordinals, there is some $K\in\mathcal{R}$ s.t. $\xi(Q)=\xi(Q/Q\cap K)$. Let us now suppose that $\xi(Q)$ is transfinite. Since $\mathcal{R}$ is a filtering family, we may find $L \leq_{\text{comp}} K\cap Q$ that is normal in Q and is open in $K\cap Q$. We can then deduce

$\to$ $\{1\}\to K\cap Q\to Q\to Q/K\cap Q\to\{1\}$

$\{1\}\to L\to K\cap Q\to K\cap Q/L\to\{1\}$

$$\xi(Q/K\cap Q)\leq\xi(Q)\leq\xi(L)+\xi(K\cap Q/L)+\xi(Q/K\cap Q)$$
$$\leq 4+\xi(Q/K\cap Q).$$

Since $\xi(Q)$ is transfinite, the only way to achieve such inequalities is if $\xi(Q/K\cap Q)=\xi(Q)$. We conclude that $\xi(Q)=\alpha+1$ for any compactly generated open subgroup Q and hence that $\xi(\mathrm{Res}(Q))\leq\alpha\Rightarrow\xi(Q)\leq\alpha+1$, completing the induction.

∎

3.3 / Regionally SIN groups

Def (SIN group)

Iff there is a basis at 1 of compact-open normal subgrps.

Def. (Regionally SIN)

If every open compactly generated subgroup is SIN.

Lemma: For a non-trivial t.d.l.c. group G, the following are equivalent:

(1) G is reginally SIN;

(2) G is elementary with $\xi(G)=2$.

↳ *Proof:*

From the definition of decomposition rank, we see that G is elementary with $\xi(G)=2$ iff every compactly generated open subgroup of G is residually discrete. By a ~~the~~ lemma we have that a compactly gen group is resi. disc. iff it is SIN, and the conclusion follows. ∎

[Obs: quasi-discrete if $QZ(G) = \{g \in G;\ C_G(g)$ is open$\}$ is dense]

Obs: Given a characteristically simple l.c.s.c. group G, then either it is quasi-discrete or has trivial quasi-center.

Also note that the class of regionally ~~SIN~~ SIN groups includes all quasi-discretes t.d.l.c.s.c. groups.

Lemma: For G a quasi-discrete l.c.s.c group, G° is central in G, and G/G° is elementary with $\xi(G/G^{\circ}) \leq 2$.

↳ Proof:

Every open subgroup contains G°, so G° centralizes $QZ(G)$. Since $QZ(G)$ is dense, G° is in fact central.

We see that G/G° is quasi-discrete, so we may assume $G^{\circ} = \{1\}$. The group G is then a t.d.l.c.s.c. group $\Rightarrow$ it is regionally SIN. Last lemma ensures that $\xi(G) \leq 2$. ■

Thrm:

If G is a non-trivial characteristically simple t.d.l.c.s.c. group then exactly one of the following holds:

(i) $\xi(G) = 2$;

(ii) $\xi(G) = \alpha + 1$ for α countable limit ordinal;

(iii) G is non-elementary.

~~Df~~ Def:

For a t.d.l.c.s.c. group G and $\alpha \geq 1$ ordinal, define the rank-α residual to be:

$$\mathrm{Res}_{\alpha}(G) := \bigcap \{ K \trianglelefteq G \mid K \text{ is closed and } \xi(G/K) \leq \alpha \}$$

The rank-α will be called finite ~~[illegible]~~ rank residual of G.

Obs: The rank-ω_1 is just the elementary residual. These rank-α residuals have many nice properties. First the rank of the quotient $G/\mathrm{Res}_{\alpha}(G)$ is restricted.

Prop: For a t.d.l.c.s.c group and α an at most countable ordinal, the following hold:

(1) $\xi(G/\mathrm{Res}_{\alpha}(G)) \leq \sup^{+}(\alpha)$;

(2) If G is comp. generated and α is a limit ordinal, then $\xi(G/\mathrm{Res}_{\alpha}(G)) < \alpha$;

(3) If G is compactly generated, then $\mathrm{Res}_{2}(G) = \mathrm{Res}(G)$.